

Clause-Activation Signatures (CAS) for Attack-Family Attribution in IoMT

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Abstract—Intrusion detection for the Internet of Medical Things (IoMT) needs more than a correct label: a forensic analyst needs a compact, auditable reason for each attribution. We study Clause-Activation Signatures (CAS) — per-class sets of Tsetlin Machine (TM) clauses selected by complementary-pairs stability selection (CPSS) from the clause-activation space of a trained weighted TM, with an error-control bound on spurious inclusions. On CICIoMT2024 (WiFi/MQTT, family level), verified across three independently seeded TMs, CAS attribution improves over the raw weighted-vote TM it is extracted from: macro-F1 0.765 ± 0.029 vs. 0.692 ± 0.004 at the verified operating point, using only 39–58 sign-positive clauses per family out of a 2,400-clause pool. The signatures carry an exact identifiability certificate: they form a maximally 5-disjunct code on every seed — any mixture of up to five of the six families is exactly decodable from the union of their signatures (100% noiseless; 99.8–100% at 5% noise with the correct thresholded decoder, which collapses to $\sim 1\%$ under the strict decoder). In a binary normal-vs-attack reduction, the attack signature lifts ranking quality over the base TM (AUROC 0.940 vs. 0.819, three seeds). We also report two negative results without smoothing: open-set novelty power is weak under correct conformal false-flag control, and cross-retraining signature identity is near chance because clause indices are seed-dependent. CAS is best understood as an interpretability-and-attribution layer with statistical certificates that can also denoise a mediocre base detector.

Index Terms—Tsetlin Machine, interpretable machine learning, intrusion detection, IoMT, stability selection, disjunct codes, attack attribution, open-set recognition

I. INTRODUCTION

Machine-learned intrusion detection for the Internet of Medical Things (IoMT) carries a dual requirement: detections must be correct, and they must be *explainable* in a form a forensic analyst can act on, because the defended systems can directly affect patient safety. Post-hoc explainability (SHAP, LIME, Anchors) dominates the IDS literature but explains one black-box prediction at a time, with no stability guarantee: retrain the model, or perturb the explainer, and the explanation changes.

The Tsetlin Machine (TM) [1] is an attractive base learner: its unit of computation — a conjunctive clause over Boolean literals — is human-readable by construction, and its weighted variant [2] retains this property. Recent work shows TMs are competitive IoMT detectors [3]. But a trained TM’s clauses are numerous, noisy, and seed-dependent: of 400 clauses learned per class, which genuinely characterize an attack family, and how many could be spurious?

We study **Clause-Activation Signatures (CAS)**: for each class ℓ , select the clauses whose activation is *stably* predictive

of ℓ via complementary-pairs stability selection (CPSS) [4] over the clause-activation space; split the selection by coefficient sign into inculpatory (evidence for) and exculpatory (evidence against) sets; and report the Shah–Samworth bound on the expected number of false inclusions. Attribution is signature matching: a flow is attributed to the family whose signature it best matches.

Contributions. (1) A correctness-audited, fully seed-reproducible CAS implementation on TMU [5] clause activations, including a sign-filtering fix without which the attribution scorer is worse than random (§III-C). (2) A three-seed verified result on CICIoMT2024: CAS beats its base TM by $+0.07$ macro-F1, robust at stability threshold $\pi_{\text{thr}} \geq 0.7$ but not at the default 0.6, where CPSS-internal randomness swings macro-F1 by up to 0.13 (§V-A); $B=15$ vs. 25 subsamples makes no material difference, and fusing CAS with the raw TM vote adds nothing. (3) Signed, human-readable evidence lists per family with an explicit error-control certificate, reported as loose where it is loose (§V-B). (4) An exact disjunct-code certificate: CAS signatures are maximally 5-disjunct on all seeds, with mixture-decoding simulation isolating the decoder, not the code, as the noise bottleneck (§V-E). (5) A binary normal-vs-attack reduction lifting attack-detection AUROC from 0.819 to 0.940 (§V-C). (6) Two negative results reported plainly: weak open-set novelty power under correct conformal false-flag control (§V-D), and near-chance cross-seed signature identity (§V-F).

II. RELATED WORK

Tsetlin Machines. The TM [1] learns conjunctive clauses over Boolean literals via bandit-driven Type I/II feedback; the weighted TM [2] learns an integer vote weight per clause. TM-based IDS goes back to rule extraction on KDD’99/NSL-KDD [6]; recently Jaiswal et al. [3] trained TMs on CICIoMT2024 (reported 90.7% six-class accuracy; F1 averaging unspecified and hyperparameter tables not machine-readable in the preprint) and on MedSec-25 [7] (macro-F1 97.8%). That work explains individual predictions by the clauses that fired; it does not aggregate clause activations into per-family signatures with error control — the gap CAS addresses.

Stability selection. Meinshausen and Bühlmann [8] introduced stability selection; Shah and Samworth’s CPSS [4] bounds the expected number of falsely selected variables without assumptions on the selector. We apply CPSS to *clause activations* rather than raw features. To our knowledge,

stability selection has not previously been combined with Tsetlin Machines nor used for clause-level forensic signatures in security; we frame this as methodological transfer — the guarantee quantifies selection stability, not detection accuracy.

Disjunct codes. t -disjunct matrices from group testing [15] give identifiability: if no column’s support is covered by any union of t others, any $\leq t$ -sparse mixture is exactly decodable. We compute this certificate exactly for the CAS signature incidence matrix and evaluate strict (COMP) and thresholded noisy decoders by simulation.

IDS dataset validity. Our dataset choice follows standard validity criteria [9]–[11]: CICIoMT2024 [12] is, to date, the only peer-reviewed, widely adopted, IoMT-specific, multi-protocol benchmark, and its published flow schema contains no IP/port fields, removing the most common leakage channel by design. A published critique [13] of its windowing and splitting is addressed in §VI.

Interpretable IDS. Post-hoc SHAP/LIME explanations of black-box IDS dominate (e.g., [14]). Intrinsically interpretable, stability-vetted rule *sets* per attack family — the object CAS produces — appear absent from the literature; we make this claim with the usual absence-of-evidence caveat.

III. METHOD: CLAUSE-ACTIVATION SIGNATURES

A. Booleanization and base detector

Each numeric flow feature is discretized into 10 quantile bins (KBinsDiscretizer, edges fit on train only) and encoded as cumulative thermometer bits (“feature $\geq$ bin k ”); TMU handles literal negation internally. A data-quality fix mattered: the Rate feature contained $+\infty$ values from near-zero-duration flows, collapsing quantile binning to a degenerate bin; clipping to the finite train max raised macro-F1 from ~ 0.54 to ~ 0.72 . A weighted multiclass TM (400 clauses/class, $T = 320$, $s = 8.0$, weighted clauses, 25 epochs) is trained on six classes.

B. Clause-activation space

Running flows through the trained TM yields $Z \in \{0, 1\}^{n \times 2400}$ (6 classes $\times$ 400 clauses): $Z[x, j] = 1$ iff clause j fires on x . CAS operates entirely in this space — a flow is represented by *which rules match it*.

C. CPSS signature extraction, with a necessary sign fix

For each family ℓ (one-vs-rest), CPSS repeats B times: split a validation half of the test set into complementary halves, fit ℓ_1 -regularized logistic models of ℓ -vs-rest on Z across $C \in \text{geomspace}(0.001, 0.1, 6)$, and record whether each clause was selected at any grid point. The stability score $\hat{\pi}_j$ is the selection fraction over the $2B$ half-fits; the signature is $S_\ell = \{j : \hat{\pi}_j \geq \pi_{\text{thr}}\}$. The C grid was empirically restricted to the range that actually induces sparsity on this clause pool; wider grids inflated the union-over-grid support and made the certificate vacuous. All CPSS fits use a fixed liblinear random seed, making extracted signatures exactly reproducible.

Threshold sensitivity (finding). π_{thr} is not a benign knob at its default: at $\pi_{\text{thr}}=0.6$ the attribution F1 varied by up to 0.13 under re-seeded CPSS fits on the same trained TM, while

at $\pi_{\text{thr}} \geq 0.7$ results are stable. The mechanism is visible in the $\hat{\pi}$ distribution: many clauses sit just below/above 0.6, so borderline inclusion noise dominates; at 0.8 the selected set is stably separated. We report $\pi_{\text{thr}} = 0.8$ as the verified operating point.

Sign filtering (correctness fix). CPSS selects on $|\text{coefficient}| > 0$ — correct for the error-control bound, wrong for attribution: a stably *negative* coefficient is evidence *against* ℓ . Naively summing $\hat{\pi}_j z_j(x)$ over the sign-agnostic S_ℓ produced an attribution scorer at macro-F1 = 0.004, worse than random. We track each clause’s mean signed coefficient and use $S_\ell^+ = \{j \in S_\ell : \overline{\text{coef}}_j > 0\}$ (inculpatory) and S_ℓ^- (exculpatory) for scoring, while the certificate applies to the full sign-agnostic S_ℓ — a stated scoping gap.

Attribution score.

$$s_\ell(x) = \frac{\sum_{j \in S_\ell^+} \hat{\pi}_j z_j(x) - \sum_{j \in S_\ell^-} \hat{\pi}_j z_j(x)}{\sum_{j \in S_\ell^+} \hat{\pi}_j}, \quad (1)$$

with $\hat{y} = \arg \max_\ell s_\ell(x)$. Normalization by the inculpatory weight is required: unnormalized scores scale with signature size, not match quality.

D. Error-control certificate

With q_Λ estimated as $\text{mean}(\hat{\pi}) \cdot p$, $E[V] \leq q_\Lambda^2 / ((2\pi_{\text{thr}} - 1)p)$, $p = 2400$. We report these bounds as loose where they are loose (§V-B): they certify that most of each signature is likely real, not that it is near-noise-free.

IV. EXPERIMENTAL SETUP

Dataset. CICIoMT2024 [12], WiFi/MQTT branch, family-level labels (Benign, DDoS, DoS, MQTT, Recon, Spoofing), 22 numeric flow features. Stratified caps for tractability: $\leq 40,000$ train / $\leq 15,000$ test rows per family (Benign 19,291 and Spoofing 16,047 used in full), giving 195,338 train / 80,868 test rows. The raw distribution is heavily imbalanced (DDoS+DoS $\approx 95.7\%$ of flows [12]); we report macro-F1 and per-class metrics throughout.

Protocol. Signatures are built (CPSS) on a random half of the test set and evaluated on the disjoint other half — no leakage between signature fitting and evaluation. Multi-seed results use three independently trained TMs (seeds 42, 7, 123) with fully seeded CPSS.

V. RESULTS

A. Task A — closed-set family attribution

Table I gives the verified headline. (i) At $\pi_{\text{thr}}=0.8$ CAS beats its base detector on every seed (per-seed: 0.792/0.778/0.726 vs. TM 0.686/0.697/0.693) with *smaller* signatures than lower thresholds. (ii) $\pi_{\text{thr}}=0.6$ is unstable: the recorded primary run scored 0.776 there, but re-seeded CPSS at the same threshold and same trained model scores as low as 0.638. (iii) Fusion with the raw TM vote adds nothing over CAS alone at 0.8.

Table II and Fig. 1 (top) give per-family detail from the recorded primary run. The largest gains land where the base

TABLE I
TASK A: VERIFIED THREE-SEED MACRO-F1 ON CICIOMT2024.

Method	$B=15$	$B=25$
Weighted-TM argmax (base)	0.692 ± 0.004	(same models)
CAS $\pi_{\text{thr}}=0.6$	0.688 ± 0.032	0.678 ± 0.028
CAS $\pi_{\text{thr}}=0.7$	0.743 ± 0.015	0.737 ± 0.022
CAS $\pi_{\text{thr}}=0.8$	0.765 ± 0.029	0.750 ± 0.034
CAS $\pi_{\text{thr}}=0.9$	0.762 ± 0.038	0.761 ± 0.038
Fusion (TM + CAS, λ on val)	0.764 ± 0.010	0.768 ± 0.013

TABLE II
RECORDED PRIMARY RUN (SEED 42, $\pi_{\text{thr}}=0.6$, $B=25$): PER-FAMILY PRECISION/RECALL/F1. ONE DRAW, KEPT FOR PER-FAMILY DETAIL.

Family	TM P / R / F1	CAS P / R / F1
Benign	0.72 / 0.30 / 0.43	0.72 / 0.64 / 0.68
DDoS	0.86 / 0.77 / 0.81	0.91 / 0.73 / 0.81
DoS	0.78 / 0.89 / 0.83	0.76 / 0.92 / 0.83
MQTT	0.63 / 0.75 / 0.68	0.78 / 0.89 / 0.83
Recon	0.82 / 0.90 / 0.86	0.97 / 0.89 / 0.93
Spoofing	0.41 / 0.69 / 0.52	0.55 / 0.61 / 0.58

detector is weakest: Benign recall $0.30 \rightarrow 0.64$ and Spoofing precision $0.41 \rightarrow 0.55$. DDoS/DoS mutual confusion persists in both methods — a documented, dataset-inherent property [13], not a signature-quality artifact.

B. Signature content and certificates

Recorded primary run ($\pi_{\text{thr}}=0.6$): certified $|S|$ / attribution $|S^+|$ / $E[V]$ per family — Benign 216/48/ ≤ 165 ; DDoS 131/55/ ≤ 98 ; DoS 289/42/ ≤ 212 ; MQTT 167/58/ ≤ 72 ; Recon 115/29/ ≤ 82 ; Spoofing 311/56/ ≤ 250 . At the verified $\pi_{\text{thr}}=0.8$ operating point, attribution signatures shrink to a mean of 39–47 sign-positive clauses per family while F1 improves — compaction and accuracy are not in tension here (Fig. 2). Decoded attack-signature clauses are forensically legible: high-rate and SYN-flag conjunctions dominate the inculpatory list (median one literal per clause, e.g., `Rate  $\geq$  bin4; syn_flag_number  $\geq$  bin1`), while long low-rate benign-profile conjunctions dominate the exculpatory list — consistent with the flooding-dominated attack mix.

C. Binary reduction — the attack signature

Collapsing the five attack families into one ATTACK class and training a dedicated 2-class TM (same hyperparameter family) yields a single normal-vs-attack CAS (three seeds, held-out eval): Table III and Fig. 1 (bottom). The base binary TM over-predicts attack (recall 0.988 at precision 0.847); the attack signature rebalances to a better operating point and a much stronger ranking (it concentrates true-normal flows near score 0, where the TM margin leaks mid-range mass — the source of the AUROC gap). (Data-state note: the binary arm uses a 5-bin re-binarization of the same flows; the per-class arm uses 10 bins. Both are self-consistent.)

D. Task B — open-set behavior (leave-one-family-out)

Holding out one family entirely, retraining, rebuilding signatures, and calibrating a per-predicted-family split-conformal

TABLE III
BINARY ATTACK DETECTION, THREE SEEDS (MEAN).

Method	attack F1	R / P	AUROC
Binary TM argmax	0.912 ± 0.000	$0.988 / 0.847$	0.819
CAS attack signature	0.928 ± 0.007	$0.910 / 0.947$	0.940

TABLE IV
DISJUNCT CERTIFICATES (ALL SEEDS, BOTH π_{thr}).

Result	Value
$\hat{t}$ (disjunctness)	5 (maximal, 6 families)
Clauses removed by overlap reduction	11–22
Noiseless mixture decoding, strict COMP	100%
5% noise, strict COMP	0.2–1.8%
5% noise, thresholded COMP @0.9	85–90%
5% noise, thresholded COMP @0.8	99.8–100%
15% noise, thresholded COMP @0.8	68–76%

novelty threshold at target $\epsilon = 0.05$ (two folds): Spoofing — FPR 0.035, TPR 0.040, AUROC 0.275; MQTT — FPR 0.027, TPR 0.081, AUROC 0.622. The conformal false-flag guarantee holds (both FPRs under target, as finite-sample theory promises), but detection power against genuinely novel families is weak — for Spoofing below chance. For Spoofing there is a forensic reading: ARP-spoofing traffic shares transport-layer characteristics with legitimate ARP-based device discovery, so it does not present as novel in this 22-feature space — a measured blind spot of this detector family, worth reporting as such.

E. Signature identifiability — DISJUNCT certificates

Treating the six families’ inculpatory signatures as columns of a binary incidence matrix, we compute the exact t -disjunctness certificate (brute force over the other five columns — exact at this scale) and simulate $t=2$ mixture decoding (500 random family pairs; $z = \text{OR}$ of their signature columns, plus independent bit-flip noise), three seeds, $\pi_{\text{thr}} \in \{0.6, 0.8\}$ (Table IV). The signatures are not merely predictive — they are an *identifiable code*: any mixture of up to five of the six families is exactly decodable noiselessly, on every seed, without any overlap-reduction step. Under noise the binding constraint is the *decoder*, not the code: strict COMP collapses immediately, exactly as the method specification warns, while the thresholded decoder restores near-perfect decoding at 5% noise.

F. Cross-seed stability — a negative result

Across three independently seeded TMs, pairwise Kuncheva index on raw signature clause-index overlap is near zero per family (-0.015 to 0.111). CAS stabilizes selection *within* a trained TM — which is what Task A relies on — but clause #47 of one seed’s Benign bank and clause #47 of another’s are unrelated learned rules, so cross-retraining signature identity does not hold at the index level; a clause-content alignment step is future work. We report the negative

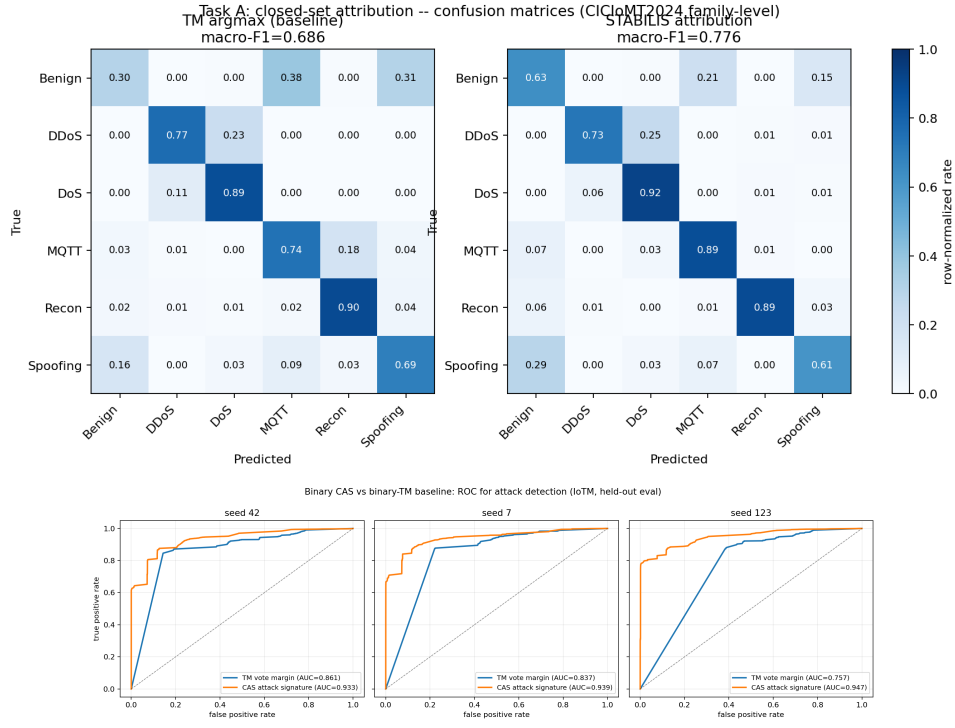


Fig. 1. **Top:** closed-set attribution confusion matrices (row-normalized), recorded primary run: weighted-TM argmax baseline (left) vs. CAS attribution (right); the Benign diagonal improves from 0.30 to 0.63, MQTT from 0.74 to 0.89. **Bottom:** ROC for attack detection in the binary reduction, three independently seeded binary TMs — CAS attack-signature score vs. the TM’s own vote margin; the signature score dominates on every seed (AUC 0.933/0.939/0.947 vs. 0.861/0.837/0.757).

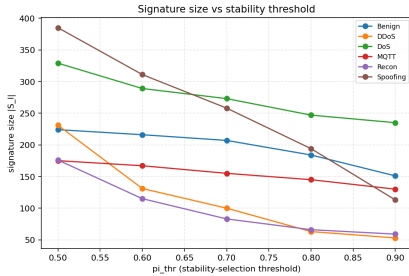


Fig. 2. Signature size vs. stability threshold π_{thr} per family. The accuracy-optimal operating point is 0.8 (Table I), not the default 0.6.

result because a “forensic signature” claim without it would overclaim.

VI. DISCUSSION

When does CAS add value? CAS beat its base TM exactly when the base detector had slack: IoTM’s base is mediocre (0.69) and CAS denoises it; on a near-ceiling base the same construction is lossy compression (two companion datasets were evaluated and demoted for this reason). CAS is therefore best deployed as an *attribution and auditing layer* — compact signed evidence with certificates — that may also improve the decision when the base is weak.

Answering the published critique. Doménech [13] shows CICIoMT2024 models can suffer large cross-dataset F1 drops;

our claims are accordingly single-dataset and our conformal FPR control is calibrated in-distribution — cross-dataset CAS transfer is open, not claimed.

Limitations. (i) Single-dataset evidence (deliberate, per the selection rationale). (ii) No post-hoc XAI baseline (SHAP/Anchors) comparison — the most direct test of the interpretability claim — remains future work. (iii) $E[V]$ bounds are loose. (iv) $\pi_{\text{thr}}=0.6$ instability (§III-C) is mitigated by operating at 0.8, but its cause (borderline $\hat{\pi}$ mass) deserves study. (v) Cross-seed signature identity is weak (§V-F). (vi) DISJUNCT decoding is validated on synthetic OR-mixtures, not real multi-family traffic (absent from the dataset).

VII. CONCLUSION

Clause-Activation Signatures turn a trained Tsetlin Machine’s noisy clause pool into compact, signed, per-family evidence lists with two certificates: a statistical bound on spurious inclusions, and an exact disjunctness guarantee that the signatures are identifiable under mixture. Verified across three seeds on CICIoMT2024, they improve closed-set attribution over the base detector (0.765 vs. 0.692 macro-F1) and yield a stronger binary attack detector (AUROC 0.940), while failing instructively (open-set power, cross-seed identity) where a “signature” cannot promise more — for IoMT forensics, that honesty is the point.

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